



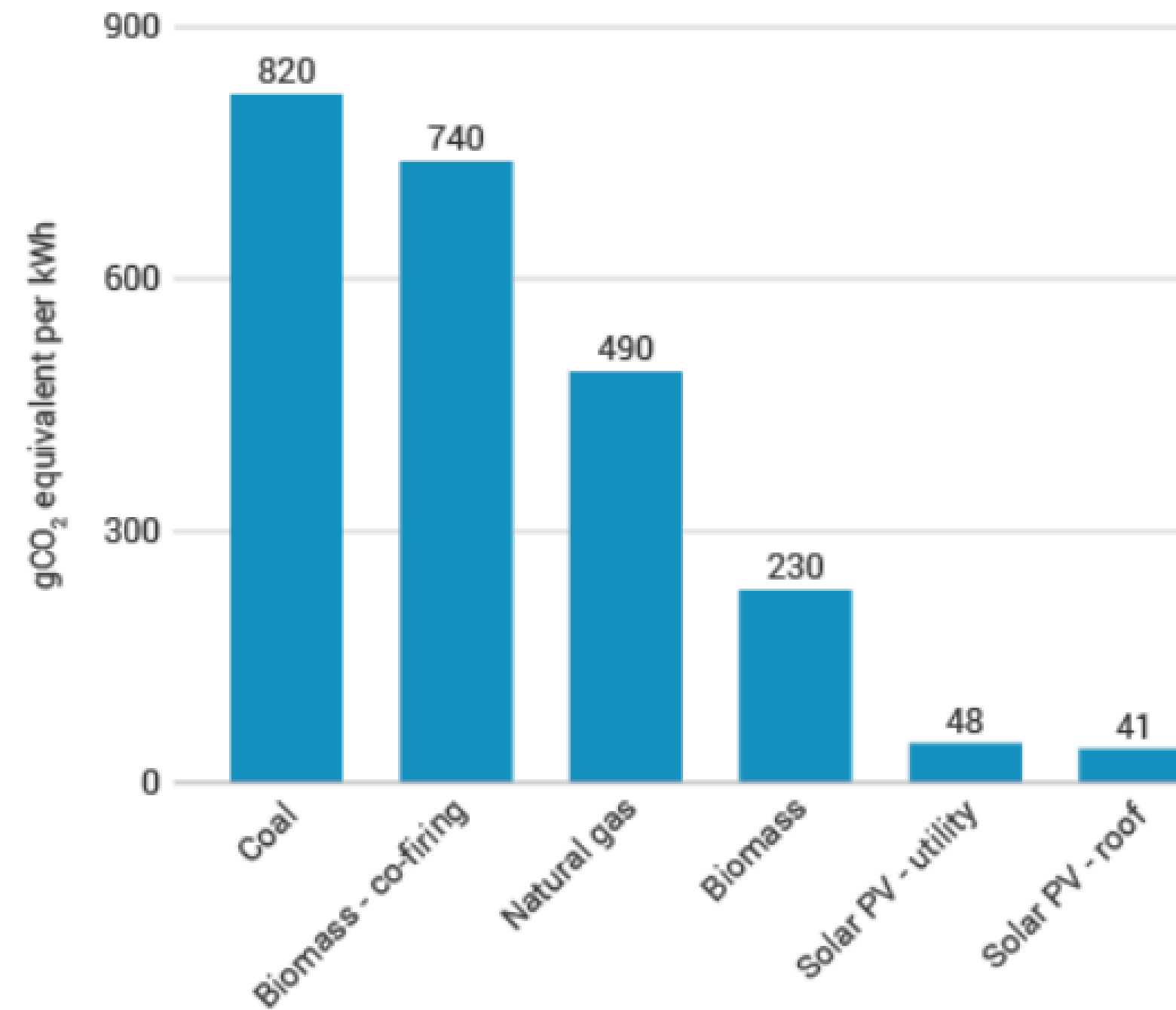
Determining the optimal location of wind turbines for a wind power plant

Carbon footprint of energy industry



Energy industry causing the most CO₂ waste because of relation to fossil fuels.

Its heavily related on fossil usage for production of energy, but in last years there is a strong tendency to move from this type of energy to more ecologically clean resources.



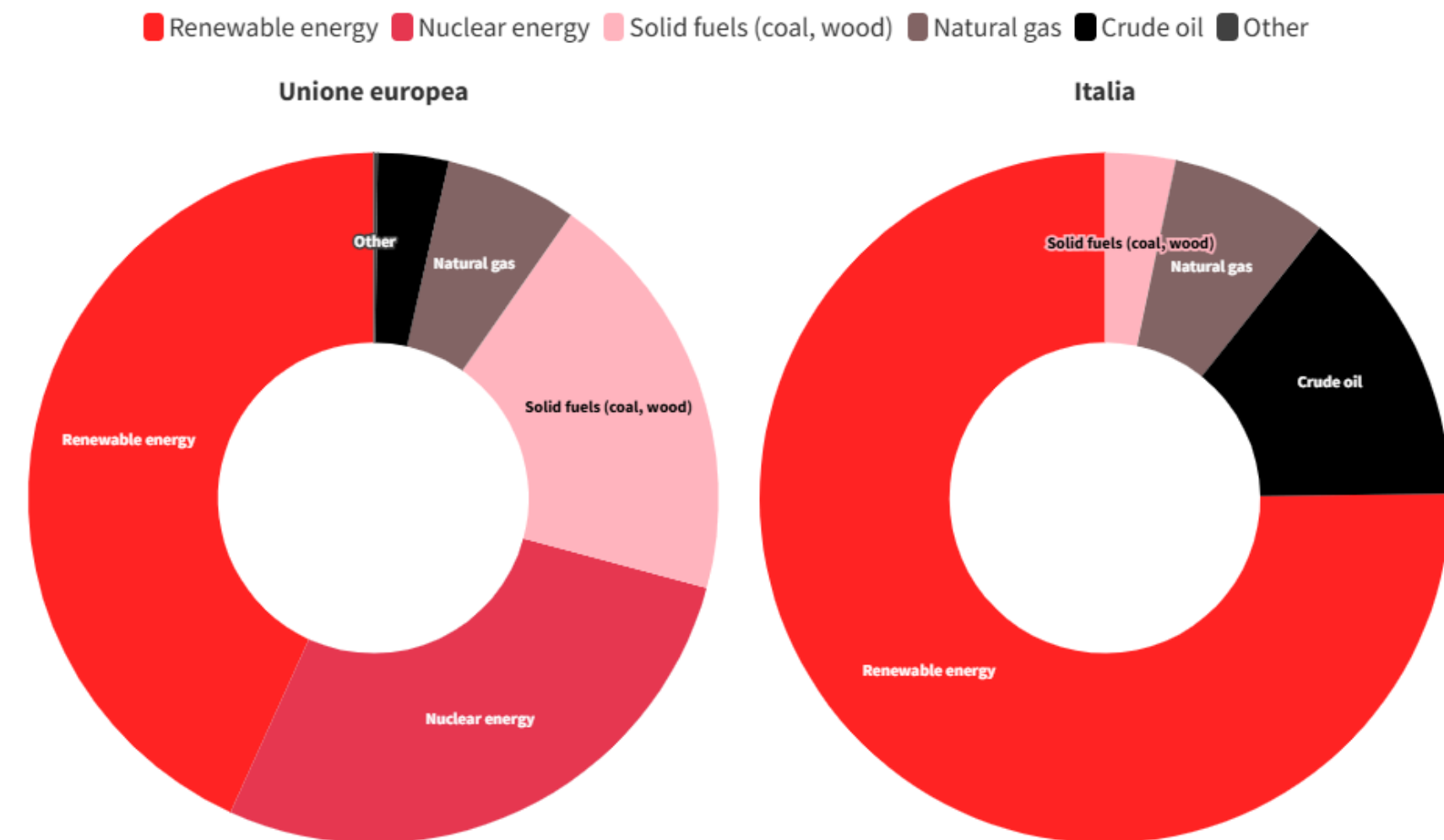
Production of renewable energy nowadays



Renewable energy had rapidly grown in Europe since last decade and significant part of overall consumed energy was produced by renewable (green) sources of energy.

In Italy production of green energy have invreased and in 2014 around 40% of national consumption of energy was fulfilled with energy, that was produce with renewable resources.

Energy produced in the EU and Italy
The % distribution by type of source



Source: [Eurostat](#)

eu news

Wind power plants statistic



Wind power plants, in particular, have produced around 7% of all electricity and 2% of all energy in the world in 2022.

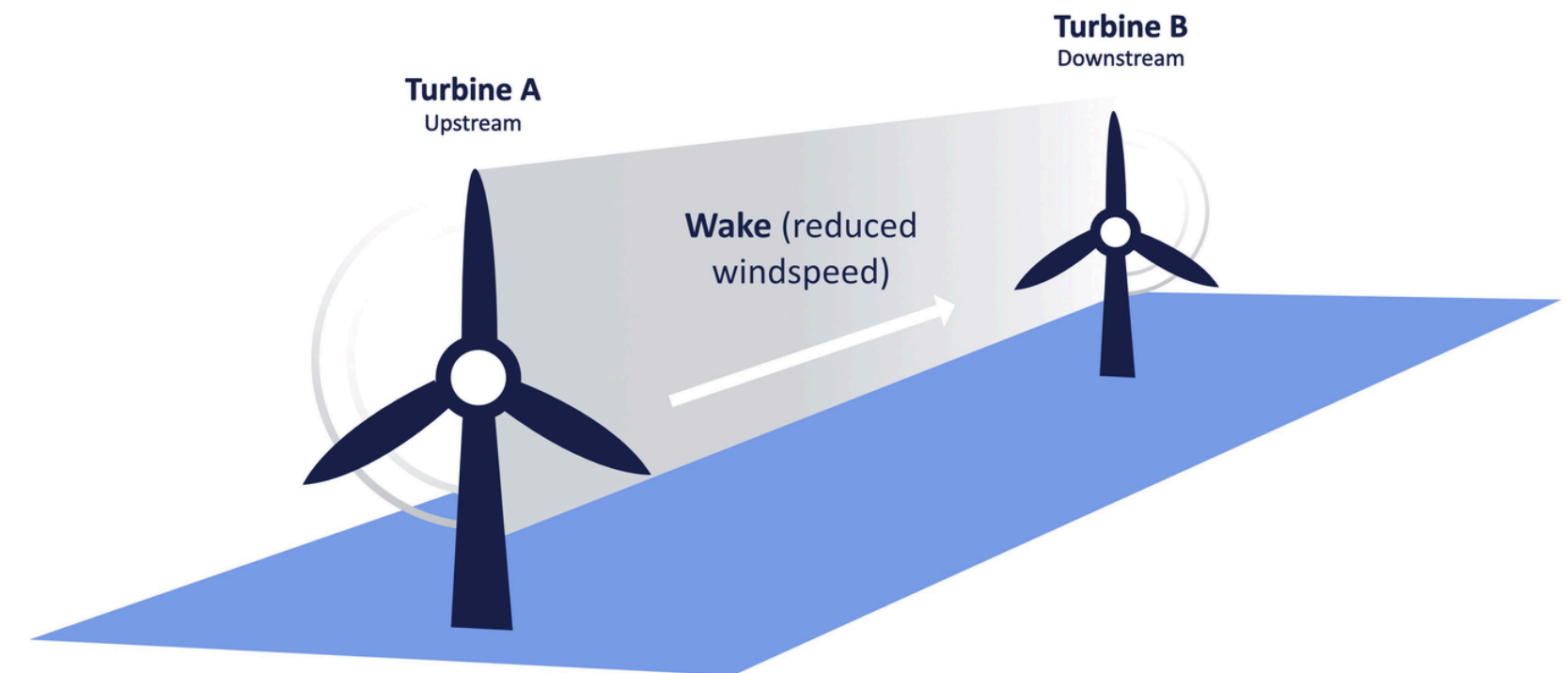
Wind power plants ones of the cheapest among all the types of plants in their inner cost and have the lowest impact on the environment.



Placement and wake effect



Wake effect can be accurate when rotation of one turbine can affect windflow that will be received by another wind turbine. This effect will lead to decreasing of power generation of affected turbines, which located in downstream.



Used data



For creating the project, as a basis was taken Kelmarsh wind farm(UK).

Main details:

- Commissioning year: 2016
- Nameplate capacity: 12 MW
- Location: West Northamptonshire, East Midlands, England, United Kingdom

Technical approach



Implemented knowledge from courses:

Machine learning and Deep learning - for creation and training neural network for necessary calculations.

Artificial intelligence - for processing the results of neural network and making informed decision, with efficiency as a main priority.

Thank you for your attention

